

Non-fragile synchronization of memristive BAM networks with random feedback gain fluctuations



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ABSTRACT

This paper investigates the non-fragile synchronization for bidirectional associative memory (BAM) neural networks with time-varying delays. A memristor-based BAM neural network is formulated and sufficient conditions are derived to guarantee its synchronization based on master–slave system approach. The non-fragile observer based feedback controller gains are assumed to have the random fluctuations, two different types of uncertainties which perturb the gains are taken into account. The Lyapunov–Krasovskii stability theory together with linear matrix inequality (LMI) approach is used to derive the delay-dependent criteria to ensure the asymptotic stability of the error system, which guarantees the master system synchronize with the slave system. The expressions for the non-fragile controller can be obtained by solving a set of LMIs using standard MATLAB toolboxes. Finally, numerical example on the chaotic system is presented to illustrate the effectiveness of the theoretical results.

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1. Introduction

Bidirectional associative memory neural networks [1] have successful application in the field of pattern recognition and artificial intelligence due to its generalization of the single-layer auto associative Hebbian correlation to a two layer pattern matched hetero-associative circuit. This class of neural network model can be implemented in a circuit where the self-feedback connection weights and the connection weights are implemented by resistors. In the past few decades, there are extensive literatures on stability analysis of BAM neural networks see [2–6]. Moreover, the time delays in real time dynamical systems are unavoidably encountered in the implementation of neural networks and may cause undesirable dynamic network behaviors such as oscillation, divergence, poor performance of networks and instability. Hence, the stability analysis problem for neural networks with time delays becomes increasingly significant and some results related to this problem have recently been published [7–12]. Also, it is well known that the stability is not the unique dynamic behavior, there are many other topics like chaos, oscillation, bifurcation and so on has to be investigated for dynamic neural networks.

The memristor is the fourth basic passive nonlinear circuit element along with resistor, inductor, and capacitor found by Chua [13]. Further, it was later identified and used to build the prototype of the nanometer size by a group of scientists from Hewlett-Packard company in 2008 [14,15]. The memristor was used to model the devices based on varied and mutually unrelated

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physical principles. It establishes a link between two physical quantities, which are in the given physical field of study analogous to the flux and the charge. So many studies proved that, using memristor device we can model a variety of systems differing in their physical nature such as mechanical, hydraulic, neuromorphic, or other systems. The investigations also discussed that, the curve of voltage against current applied to the memristor shows pinched hysteresis loop, these pinched hysteresis loop serves as the fingerprint for memristor. Its value, called memristance (or memductance), is not unique, since it depends on the magnitude and polarity of the voltage applied to it and the length of the time that the voltage has been applied [13]. Because of the varies features explained above, recently, dynamical behaviors of memristor-based recurrent neural networks (MRNNs) has been attracted increasing attention of researchers [12,16–18]. Furthermore, the anti-synchronization control of a class of MRNNs has been covered in [17] and [12] addresses exponential synchronization of MRNNs with time delays. Recently, the investigations shows that, if we replace the resistors with memristors then we can build a new class of BAM neural networks model called memristor-based BAM neural networks. Therefore, it has a wider significance to study the dynamical behaviors of these kinds of BAM models. Recently, some complex dynamical properties of memristor-based time-varying delayed BAM neural networks with periodic or almost periodic inputs has been studied in [19].

On the other hand, the synchronization control of chaotic systems plays an important role in applications such as secure communication, image encryption, image segmentation, two-dimensional motion control and information processing [20–22]. Chaotic synchronization means the states of the connected systems are coincide each other, that is, it is a method to synchronize two identical chaotic systems (drive-response concept) with different initial conditions. Chaotic signal is conjectured to be used in communication schemes because of its inherent wide band characteristic, as a result, many synchronization methods work has been carried out for chaotic neural networks with time delays [23–30]. The problem of synchronization of chaotic systems with randomly occurring uncertainties via stochastic sampled-data control has been investigated in [31]. In [32], Tang and Wong studied the distributed synchronization and pinning distributed synchronization of stochastic coupled neural networks via randomly occurring control. The synchronizability of coupled memristive neural networks in the presence of state-dependent switching and time delay are established in [33]. The problem of non-fragile synchronization control for complex networks with time-varying coupling delay and missing data has been discussed in [34] and delay-dependent synchronization criteria have been derived and the design method of non-fragile controller has been given in [35]. However, up to now, there is no result on non-fragile synchronization of MBAMNNs with time-varying delays.

However, it is well known that the designed controller should be able to tolerate certain degree of plant parameter uncertainties as well as controller gain uncertainties and fluctuations due to analog to digital or digital to analog conversion and finite word lengths in numerical computations, in addition to the parameter-tuning margins. Researches have shown that the small perturbations in control parameters could even destabilize the closed-loop system [36]. Therefore, it is significantly important that any controller design should be able to tackle some level of fluctuations or variations in its own gains. Hence, it is useful to study the synchronization with the help of non-fragile controllers that are very accurate with respect to both plant uncertainties and controller uncertainties. The models of additive and multiplicative uncertainties are used to describe the controller gain variation in [37]. Recently, the problems of robust non-fragile control for uncertain discrete-delay large-scale systems under two classes of controller gain variations are considered in [38].

Motivated by the aforementioned discussion, in this work, we deal with the non-fragile synchronization of MBAMNNs with time varying delays and gain fluctuations. It is assumed that the controller gain fluctuations appear in the random ways and obeys certain Bernoulli distributed white noise sequences and moreover, two types of perturbations are considered in the main proof. Sufficient conditions to guarantee the slave system synchronizes with the master system are derived by applying the master–slave synchronization concept, LMIs analysis and Lyapunov stability theory. Also, the desired non-fragile gain controllers can be obtained by solving a set of LMIs. Finally, numerical examples have been demonstrated to show the potentiality of the developed theoretical results.

The structure of the paper is organized as follows. In the following section, the model formulation and some preliminaries of the memristive based BAMNNs are presented. Based on LMI approach and Lyapunov stability theory, Section 3 develops two different kinds of non-fragile synchronization for the considered system. In Section 4, a numerical example is presented to demonstrate the validity of the main results. Finally, conclusion is summarized in Section 5.

2. Problem description and preliminaries

In this section, we construct a general class of BAMNNs from the aspects of circuit theory and memristor physical properties, which reproduce the characteristic time hysteresis behavior of memristor devices. First, we consider a general class of BAM neural networks based on the Kirchhoff current law, the i -th and j -th subsystems are described by the following system of equations:

$$\begin{cases} \dot{x}_i(t) = -x_i(t) + \frac{1}{\tilde{C}_i} \sum_{j=1}^m \frac{\tilde{f}_j(y_j(t))}{\mathcal{R}_{ji}} \times \text{sgn}_{ji} + \frac{1}{\tilde{C}_i} \sum_{j=1}^m \frac{\tilde{f}_j(y_j(t - \tau(t)))}{\mathcal{F}_{ji}} \times \text{sgn}_{ji} \\ \quad + \frac{\mathcal{I}_i(t)}{\tilde{C}_i}, i = 1, 2, \dots, n, \\ \dot{y}_j(t) = -y_j(t) + \frac{1}{\tilde{C}_j} \sum_{i=1}^n \frac{\tilde{g}_i(x_i(t))}{\tilde{\mathcal{R}}_{ij}} \times \text{sgn}_{ij} + \frac{1}{\tilde{C}_j} \sum_{i=1}^n \frac{\tilde{g}_i(x_i(t - \tau(t)))}{\tilde{\mathcal{F}}_{ij}} \times \text{sgn}_{ij} \\ \quad + \frac{\mathcal{J}_j(t)}{\tilde{C}_j}, t \geq 0, j = 1, 2, \dots, m. \end{cases} \quad (1)$$

where $\text{sgin}_{ji} = \text{sgin}_{ij} = 1$ for $i \neq j$ and -1 for $i = j$; $x_i(t)$ is the voltage of the capacitor C_i ; $y_j(t)$ is the voltage of the capacitor $\tilde{C}_j$; $\mathcal{R}_{ji}$ denotes the resistor between the continuous feedback function $\tilde{f}_j(y_j(t))$ and $x_i(t)$; $\mathcal{F}_{ji}$ denotes the resistor between the continuous feedback function $\tilde{f}_j(y_j(t - \tau(t)))$ and $x_i(t)$; $\tilde{\mathcal{R}}_{ij}$ denotes the resistor between the continuous feedback function $\tilde{g}_i(x_i(t))$ and $y_j(t)$; $\tilde{\mathcal{F}}_{ij}$ denotes the resistor between the continuous feedback function $\tilde{g}_i(x_i(t - \tau(t)))$ and $y_j(t)$; $\mathcal{I}_i(t)$ denotes the continuous external input or bias function on the i -th unit at time t ; $\mathcal{J}_j(t)$ denotes the continuous external input or bias function on the j -th unit and the non negative continuous variable.

It is noted that the time delays are often encountered in neural networks in two ways, one way is due to the signal propagation between neurons in network, another way is because autapse effect that time delay finds in a close loop, which neuron connects to itself via synapse. In electronic implementations of neural networks, time-delay is frequently encountered owing to the finite switching speed of the amplifiers and the inherent transmission time between neurons such as signal transmission delay, signal processing and propagation delay due to integration and communication which are ubiquitous. In this paper the $\tau(t)$ represents the transmission time-varying delay and satisfies $0 \leq \tau(t) \leq \tau$, and $\dot{\tau}(t) \leq \mu$, where μ is known constant. The neural networks with delay due to autapse effects can be found in [39–42]. The feedback functions satisfies the following assumption: **(A1)** For $i = \{1, 2, \dots, n\}$ and $j = \{1, 2, \dots, m\}$, the activation functions $\tilde{f}_j$ and $\tilde{g}_i$ are Lipschitz continuous and for all x_1, x_2, y_1 and $y_2 \in \mathbb{R}$, there exists a real number e_j and l_i such that

$$0 \leq \frac{\tilde{f}_j(y_1) - \tilde{f}_j(y_2)}{y_1 - y_2} \leq s_j, \quad \tilde{f}_j(0) = 0, \quad y_1 \neq y_2, \quad 0 \leq \frac{\tilde{g}_i(x_1) - \tilde{g}_i(x_2)}{x_1 - x_2} \leq l_i, \quad \tilde{g}_i(0) = 0, \quad x_1 \neq x_2.$$

Then, it follows from (1) that

$$\begin{cases} \dot{x}_i(t) = -x_i(t) + \sum_{j=1}^m a_{ji}^* \tilde{f}_j(y_j(t)) + \sum_{j=1}^m b_{ji}^* \tilde{f}_j(y_j(t - \tau(t))) + I_i(t), \\ \dot{y}_j(t) = -y_j(t) + \sum_{i=1}^n c_{ij}^* \tilde{g}_i(x_i(t)) + \sum_{i=1}^n d_{ij}^* \tilde{g}_i(x_i(t - \tau(t))) + J_j(t). \end{cases} \quad (2)$$

where $a_{ji}^* = \frac{\text{sgin}_{ji}}{\mathcal{C}_i \mathcal{R}_{ji}}$, $b_{ji}^* = \frac{\text{sgin}_{ji}}{\mathcal{C}_i \mathcal{F}_{ji}}$, $c_{ij}^* = \frac{\text{sgin}_{ij}}{\tilde{\mathcal{C}}_j \tilde{\mathcal{R}}_{ij}}$, $d_{ij}^* = \frac{\text{sgin}_{ij}}{\tilde{\mathcal{C}}_j \tilde{\mathcal{F}}_{ij}}$, $I_i(t) = \frac{\mathcal{I}_i(t)}{\mathcal{C}_i}$, $J_j(t) = \frac{\mathcal{J}_j(t)}{\tilde{\mathcal{C}}_j}$. Now, by replacing the resistors $\mathcal{R}_{ji}$, $\mathcal{F}_{ji}$, $\tilde{\mathcal{R}}_{ij}$ and $\tilde{\mathcal{F}}_{ij}$ in the primitive BAM neural networks (1) or (2) with memristors, whose memductances are $\mathcal{W}_{ji}$, $\mathcal{M}_{ji}$, $\tilde{\mathcal{W}}_{ij}$ and $\tilde{\mathcal{M}}_{ij}$, respectively, then we can construct the memristor-based BAM neural networks of the following form:

$$\begin{cases} \dot{x}_i(t) = -x_i(t) + \sum_{j=1}^m a_{ji}(x_i(t)) \tilde{f}_j(y_j(t)) + \sum_{j=1}^m b_{ji}(x_i(t)) \tilde{f}_j(y_j(t - \tau(t))) + I_i(t), \\ \dot{y}_j(t) = -y_j(t) + \sum_{i=1}^n c_{ij}(y_j(t)) \tilde{g}_i(x_i(t)) + \sum_{i=1}^n d_{ij}(y_j(t)) \tilde{g}_i(x_i(t - \tau(t))) + J_j(t), \end{cases} \quad (3)$$

where $a_{ji}(x_i(t)) = \frac{\mathcal{W}_{ji}}{\mathcal{C}_i} \text{sgin}_{ji}$, $b_{ji}(x_i(t)) = \frac{\mathcal{M}_{ji}}{\mathcal{C}_i} \text{sgin}_{ji}$, $c_{ij}(y_j(t)) = \frac{\tilde{\mathcal{W}}_{ij}}{\tilde{\mathcal{C}}_j} \text{sgin}_{ij}$, $d_{ij}(y_j(t)) = \frac{\tilde{\mathcal{M}}_{ij}}{\tilde{\mathcal{C}}_j} \text{sgin}_{ij}$ and

$$\begin{aligned} a_{ji}(x_i(t)) &= \begin{cases} \hat{a}_{ji}, & |x_i(t)| < \Lambda_i, \\ \check{a}_{ji}, & |x_i(t)| > \Lambda_i, \end{cases} & b_{ji}(x_i(t)) &= \begin{cases} \hat{b}_{ji}, & |x_i(t)| < \Lambda_i, \\ \check{b}_{ji}, & |x_i(t)| > \Lambda_i, \end{cases} \\ c_{ij}(y_j(t)) &= \begin{cases} \hat{c}_{ij}, & |y_j(t)| < \tilde{\Lambda}_j, \\ \check{c}_{ij}, & |y_j(t)| > \tilde{\Lambda}_j, \end{cases} & d_{ij}(y_j(t)) &= \begin{cases} \hat{d}_{ij}, & |y_j(t)| < \tilde{\Lambda}_j, \\ \check{d}_{ij}, & |y_j(t)| > \tilde{\Lambda}_j, \end{cases} \end{aligned}$$

where the switching jumps Λ_i and $\tilde{\Lambda}_j$ are all non-negative, a_{ji} , b_{ji} , c_{ij} and d_{ij} are known constant numbers.

According to the analysis above, $a_{ji}(x_i(t))$, $b_{ji}(x_i(t))$, $c_{ij}(y_j(t))$ and $d_{ij}(y_j(t))$ in this system are changed according to the states $x_i(t)$, $x_i(t - \tau(t))$, $y_j(t)$ and $y_j(t - \tau(t))$ of the system, respectively. It is clear that $a_{ji}(x_i)$ (or $b_{ji}(x_i)$) and $c_{ij}(y_j)$ (or $d_{ij}(y_j)$) are discontinuous or piecewise continuous functions with at most two points of discontinuity $\pm \Lambda_i$ and $\pm \tilde{\Lambda}_j$, respectively. Meanwhile, the discontinuous functions $a_{ji}(x_i)$ (or $b_{ji}(x_i)$) and $c_{ij}(y_j)$ (or $d_{ij}(y_j)$) are undefined at the points of discontinuity $\pm \Lambda_i$ and $\pm \tilde{\Lambda}_j$, respectively. Now let $\bar{co}[\mathbb{E}]$ denote the closure of the convex hull of given set $\mathbb{E}$. Obviously, we can obtain that

$$\begin{aligned} \bar{co}[a_{ji}(x_i)] &= \begin{cases} \hat{a}_{ji}, & |x_i| < \Lambda_i \\ [\underline{a}_{ji}, \bar{a}_{ji}], & |x_i| = \Lambda_i \\ \check{a}_{ji}, & |x_i| > \Lambda_i, \end{cases} & \bar{co}[b_{ji}(x_i)] &= \begin{cases} \hat{b}_{ji}, & |x_i| < \Lambda_i \\ [\underline{b}_{ji}, \bar{b}_{ji}], & |x_i| = \Lambda_i \\ \check{b}_{ji}, & |x_i| > \Lambda_i, \end{cases} \\ \bar{co}[c_{ij}(y_j)] &= \begin{cases} \hat{c}_{ij}, & |y_j| < \tilde{\Lambda}_j \\ [\underline{c}_{ij}, \bar{c}_{ij}], & |y_j| = \tilde{\Lambda}_j \\ \check{c}_{ij}, & |y_j| > \tilde{\Lambda}_j, \end{cases} & \bar{co}[d_{ij}(y_j)] &= \begin{cases} \hat{d}_{ij}, & |y_j| < \tilde{\Lambda}_j \\ [\underline{d}_{ij}, \bar{d}_{ij}], & |y_j| = \tilde{\Lambda}_j \\ \check{d}_{ij}, & |y_j| > \tilde{\Lambda}_j, \end{cases} \end{aligned}$$

where $\bar{a}_{ji} = \max\{\hat{a}_{ji}, \check{a}_{ji}\}$, $\underline{a}_{ji} = \min\{\hat{a}_{ji}, \check{a}_{ji}\}$, $\bar{b}_{ji} = \max\{\hat{b}_{ji}, \check{b}_{ji}\}$, $\underline{b}_{ji} = \min\{\hat{b}_{ji}, \check{b}_{ji}\}$, $\bar{c}_{ij} = \max\{\hat{c}_{ij}, \check{c}_{ij}\}$, $\underline{c}_{ij} = \min\{\hat{c}_{ij}, \check{c}_{ij}\}$, $\bar{d}_{ij} = \max\{\hat{d}_{ij}, \check{d}_{ij}\}$, $\underline{d}_{ij} = \min\{\hat{d}_{ij}, \check{d}_{ij}\}$, for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. That is to say $\bar{co}[a_{ji}(x_i)]$ (or $\bar{co}[b_{ji}(x_i)]$) and $\bar{co}[c_{ij}(y_j)]$ (or $\bar{co}[d_{ij}(y_j)]$), are intervals with non-empty

interior when $a_{ji}(x_i)$ (or $b_{ji}(x_i)$) and $c_{ij}(y_j)$ (or $d_{ij}(y_j)$) are discontinuous at $|x_i| = \Lambda_i$ and $|y_j| = \tilde{\Lambda}_j$, respectively, while $\tilde{co}[a_{ji}(x_i)] = \{a_{ji}(x_i)\}$ (or $\tilde{co}[b_{ji}(x_i)]$) and $\tilde{co}[c_{ij}(y_j)]$ (or $\tilde{co}[d_{ij}(y_j)]$) are singleton sets when $a_{ji}(x_i)$ (or $b_{ji}(x_i)$) and $c_{ij}(y_j)$ (or $d_{ij}(y_j)$) are continuous at x_i (or y_j), respectively. Therefore, this BAM neural network based on memristors is a state-dependent switched system. Now from (3), we have

$$\begin{cases} \dot{x}_i(t) = -x_i(t) + \sum_{j=1}^m \tilde{co}[a_{ji}(x_i(t))]\tilde{f}_j(y_j(t)) + \sum_{j=1}^m \tilde{co}[b_{ji}(x_i(t))]\tilde{f}_j(y_j(t - \tau(t))) + I_i(t), \\ \dot{y}_j(t) = -y_j(t) + \sum_{i=1}^n \tilde{co}[c_{ij}(y_j(t))]\tilde{g}_i(x_i(t)) + \sum_{i=1}^n \tilde{co}[d_{ij}(y_j(t))]\tilde{g}_i(x_i(t - \tau(t))) + J_j(t), \end{cases} \quad (4)$$

Definition 2.1. [33] Suppose $E \subset \mathcal{R}^n$. Then $x \mapsto F(x)$ is called as a set-valued map defined on E , if for each point x of E , there corresponds a nonempty set $F(x) \subset \mathcal{R}^n$. A set valued map F with nonempty values is said to be upper semicontinuous at $x_0 \in E$ if, for any open set N containing $F(x_0)$, there exists a neighborhood M of x_0 such that $F(M) \subset N$. $F(x)$ is said to have a closed (convex, compact) image if for each $x \in E$; $F(x)$ is closed (convex, compact).

Based on Definition 2.1, set-valued maps and Filippov solution,

$$\begin{cases} \dot{x}_i(t) \in -x_i(t) + co\{\hat{A}, \check{A}\}\tilde{f}_j(y_j(t)) + co\{\hat{B}, \check{B}\}\tilde{f}_j(y_j(t - \tau(t))) + I_i(t), \\ \dot{y}_j(t) \in -y_j(t) + co\{\hat{C}, \check{C}\}\tilde{g}_i(x_i(t)) + co\{\hat{D}, \check{D}\}\tilde{g}_i(x_i(t - \tau(t))) + J_j(t), \end{cases} \quad (5)$$

holds for almost everywhere $t \in (0, +\infty)$. Moreover, $co\{\hat{A}, \check{A}\} = [\hat{A}, \check{A}]$, $co\{\hat{B}, \check{B}\} = [\hat{B}, \check{B}]$, $co\{\hat{C}, \check{C}\} = [\hat{C}, \check{C}]$ and $co\{\hat{D}, \check{D}\} = [\hat{D}, \check{D}]$. Or equivalently, there exist $A \in co\{\hat{A}, \check{A}\}$, $B \in co\{\hat{B}, \check{B}\}$, $C \in co\{\hat{C}, \check{C}\}$ and $D \in co\{\hat{D}, \check{D}\}$, such that

$$\mathcal{M} : \begin{cases} \dot{x}(t) = -x(t) + A\tilde{f}(y(t)) + B\tilde{f}(y(t - \tau(t))) + I(t), \\ \dot{y}(t) = -y(t) + C\tilde{g}(x(t)) + D\tilde{g}(x(t - \tau(t))) + J(t), \end{cases} \quad (6)$$

where $x(t) = (x_1(t), \dots, x_n(t))^T$, $y(t) = (y_1(t), \dots, y_m(t))^T$, $\tilde{f}(y(t)) = (\tilde{f}_1(y_1(t)), \dots, \tilde{f}_m(y_m(t)))^T$, $\tilde{g}(x(t)) = (\tilde{g}_1(x_1(t)), \dots, \tilde{g}_n(x_n(t)))^T$.

Here we use the master-slave synchronization approach to derive the synchronization criteria, where the master system with the state variables denoted by x_i and y_j and the slave system having identical dynamical equations denoted by the state variables $\hat{x}_i$ and $\hat{y}_j$. However, the initial conditions on the master system are different from those of the slave system. In order to derive the synchronization behavior for the considered memristive BAM neural networks, we consider the following slave system corresponding to the master system (3) or (6):

$$S : \begin{cases} \dot{\hat{x}}(t) = -\hat{x}(t) + A\tilde{f}(\hat{y}(t)) + B\tilde{f}(\hat{y}(t - \tau(t))) + I(t) + u(t), \\ \dot{\hat{y}}(t) = -\hat{y}(t) + C\tilde{g}(\hat{x}(t)) + D\tilde{g}(\hat{x}(t - \tau(t))) + J(t) + v(t), \end{cases} \quad (7)$$

where $u(t)$, $v(t)$ are the appropriate control inputs to be designed in order to obtain a certain control objectives. The initial conditions of (7) are $\hat{x}(s) = \kappa_1(s) \in \mathcal{C}([-\tau, 0], \mathbb{R}^n)$ and $\hat{y}(s) = \kappa_2(s) \in \mathcal{C}([-\tau, 0], \mathbb{R}^m)$. Let the synchronization error signals $e_x(t) = \hat{x}(t) - x(t)$, $e_y(t) = \hat{y}(t) - y(t)$, $e_x(t) = (e_{x_1}(t), \dots, e_{x_n}(t))^T$ and $e_y(t) = (e_{y_1}(t), \dots, e_{y_m}(t))^T$. Thus, error dynamics between systems (3) and (7) can be expressed as:

$$\begin{cases} \dot{e}_x(t) = -e_x(t) + Af(e_y(t)) + Bf(e_y(t - \tau(t))) + u(t), \\ \dot{e}_y(t) = -e_y(t) + Cg(e_x(t)) + Dg(e_x(t - \tau(t))) + v(t), \end{cases} \quad (8)$$

where $f(e_y(t)) = \tilde{f}(\hat{y}(t)) - \tilde{f}(y(t))$ and $g(e_x(t)) = \tilde{g}(\hat{x}(t)) - \tilde{g}(x(t))$. Now, we define the non-fragile state feedback control laws given by

$$u(t) = (K_1 + \alpha_1(t)\Delta K_1(t))e_x(t) \text{ and } v(t) = (K_2 + \alpha_2(t)\Delta K_2(t))e_y(t) \quad (9)$$

where K_1, K_2 are the controller gain matrices to be determined; the real-valued matrices $\Delta K_1(t), \Delta K_2(t)$ represent the possible gain fluctuations; $\alpha_1(t), \alpha_2(t) \in \mathbb{R}$ are stochastic variables, which are introduced to describe the phenomena of randomly occurring controller gain fluctuations. Further, $\alpha_1(t), \alpha_2(t)$ are Bernoulli-distributed white noise sequences taking on values in $(0, 1)$ with $\Pr\{\alpha_1(t) = 1\} = \alpha_a$, $\Pr\{\alpha_1(t) = 0\} = 1 - \alpha_a$, $\Pr\{\alpha_2(t) = 1\} = \alpha_b$, $\Pr\{\alpha_2(t) = 0\} = 1 - \alpha_b$, where $\alpha_a, \alpha_b \in [0, 1]$ are known constants.

In this work, we assumed that $\Delta K_1(t)$ and $\Delta K_2(t)$ has two different types of structures given by:

Type 1. Perturbation is independent on matrices K_1 and K_2 (Additive gain variations):

$$\Delta K_1(t) = H_1 F(t) M_1, \quad \Delta K_2(t) = H_2 F(t) M_2.$$

Type 2. Perturbation is dependent on matrices K_1 and K_2 (Multiplicative gain variations):

$$\Delta K_1(t) = H_1 F(t) M_1 K_1, \quad \Delta K_2(t) = H_2 F(t) M_2 K_2,$$

where H_1, M_1, H_2, M_2 are known constant matrices with appropriate dimensions and $F(t)$ is an unknown time-varying matrix with Lebesgue measurable elements bounded by $F^T(t)F(t) \leq I$. Now, we are in the position to write the closed-loop error system given by:

$$\mathcal{E} : \begin{cases} \dot{e}_x(t) = Af(e_y(t)) + Bf(e_y(t - \tau(t))) + (K_1 + \alpha_1(t)\Delta K_1(t) - I)e_x(t), \\ \dot{e}_y(t) = Cg(e_x(t)) + Dg(e_x(t - \tau(t))) + (K_2 + \alpha_2(t)\Delta K_2(t) - I)e_y(t). \end{cases} \quad (10)$$

Lemma 2.2 (Jensen's Inequality). For any matrix $R \in \mathcal{R}^{n \times n}$, $R = R^T > 0$, scalars $a < b$, vector $\omega : [a, b] \mapsto \mathcal{R}^n$ such that the integration concerned is well defined, then

$$\begin{aligned} (b-a) \int_a^b \omega^T(s)R\omega(s)ds &\geq \left(\int_a^b \omega(s)ds \right)^T R \left(\int_a^b \omega(s)ds \right), \\ \frac{(b-a)^2}{2} \int_a^b \int_{t+\theta}^t \omega^T(s)R\omega(s)dsd\theta &\geq \left(\int_a^b \int_{t+\theta}^t \omega(s)dsd\theta \right)^T R \left(\int_a^b \int_{t+\theta}^t \omega(s)dsd\theta \right), \\ \frac{(b-a)^3}{6} \int_a^b \int_{\theta_1}^0 \int_{t+\theta_2}^t \omega^T(s)R\omega(s)dsd\theta_2d\theta_1 &\geq \left(\int_a^b \int_{\theta_1}^0 \int_{t+\theta_2}^t \omega(s)dsd\theta_2d\theta_1 \right)^T R \left(\int_a^b \int_{\theta_1}^0 \int_{t+\theta_2}^t \omega(s)dsd\theta_2d\theta_1 \right). \end{aligned}$$

2.1. Notations

Throughout this paper, solutions of the systems considered in the following are intended in the Filippov's sense. $\mathbb{R}^n$ denotes the n -dimensional space with the scalar product $\langle \cdot, \cdot \rangle$ or real matrices $\tilde{\Pi}$ and $\hat{\Pi}$. Let $\tilde{a}_{ij} = \max \{ \hat{a}_{ij}, \check{a}_{ij} \}$, $\underline{a}_{ij} = \min \{ \hat{a}_{ij}, \check{a}_{ij} \}$, $\tilde{b}_{ij} = \max \{ \hat{b}_{ij}, \check{b}_{ij} \}$, $\underline{b}_{ij} = \min \{ \hat{b}_{ij}, \check{b}_{ij} \}$, $\tilde{a}_{ij} = \max \{ \hat{a}_{ij} \}$, $\tilde{b}_{ij} = \max \{ \hat{b}_{ij} \}$, for $i, j = 1, 2, \dots, n$. $P > 0$ means that the matrix P is a real symmetric positive definite matrix, whereas $P < 0$ indicates a negative-definite matrix. I and 0 represent the identity matrix and zero matrix with compatible dimension. In symmetric block matrices or long matrix expressions, we use an asterisk $(*)$ to represent a term that is induced by symmetry. $\text{diag}\{\cdot\}$ stands for a block-diagonal matrix. For matrices $\mathcal{Y} = (y_{ij})_{n \times n}$, $\mathcal{Z} = (z_{ij})_{n \times n}$, $\mathcal{Y} \gg \mathcal{Z}$ ($\mathcal{Y} \ll \mathcal{Z}$) means that $y_{ij} > z_{ij}$ ($y_{ij} < z_{ij}$), for $i, j = 1, 2, \dots, n$. And by the interval matrix $[\mathcal{Y}, \mathcal{Z}]$, it follows that $\mathcal{Y} \ll \mathcal{Z}$. For $\forall \mathcal{L} = (l_{ij})_{n \times n} \in [\mathcal{Y}, \mathcal{Z}]$, it means $\mathcal{Y} \ll \mathcal{L} \ll \mathcal{Z}$, i.e., $(y_{ij})_{n \times n} < (l_{ij})_{n \times n} < (z_{ij})_{n \times n}$ for $i, j = 1, 2, \dots, n$. $\lambda_{\max}(\mathcal{Q})$, $\lambda_{\min}(\mathcal{Q})$ denote the maximum eigenvalue and minimum eigenvalue of the matrix $\mathcal{Q}$, respectively.

3. Main results

In this section, we will establish a criterion to implement the non-fragile synchronization of MBAMNNs with time-varying delays in the presence of controller gain perturbations. The following theorem presents a sufficient condition to guarantee the non-fragile synchronization for the MBAMNNs with additive gain variations (Type 1).

Theorem 3.1. Assume that the condition (A1) holds. If there exist symmetric positive definite matrices P_i , R_3 , R_4 , $R_i = \begin{bmatrix} R_{i1} & R_{i2} \\ * & R_{i3} \end{bmatrix} > 0$, $Z_1 > 0$, $Z_2 > 0$, $Z_3 > 0$, $Z_4 > 0$, diagonal matrices $G_s > 0$, $s = 1, 2, \dots, 6$, N_i , $i = 1, 2$ and scalars ϵ_1, ϵ_2 with appropriate dimensions such that the following LMI holds:

$$\begin{bmatrix} \bar{\Psi} & \bar{M}_a & \bar{M}_c \\ * & -\epsilon_1 I & 0 \\ * & * & -\epsilon_2 I \end{bmatrix} < 0 \quad (11)$$

where $\bar{\Psi} = [\hat{\Psi}_{k \times l}]_{16 \times 16} + \text{diag}\{\epsilon_1 \alpha_a^2 H_1^T H_1 \quad 0_{7n \times 7n} \quad \epsilon_2 \alpha_b^2 H_2^T H_2 \quad 0_{7n \times 7n}\}$, $\bar{M}_a = [0_{n \times 3n} \quad X_1^T M_1^T \quad 0_{n \times 12n}]^T$, $\bar{M}_c = [0_{n \times 11n} \quad X_2^T M_2^T \quad 0_{n \times 4n}]^T$,

$$\begin{aligned} \hat{\Psi}_{1,1} &= \hat{R}_3 + \tau \hat{R}_{11} - \frac{1}{\tau} \hat{R}_{13} - 2\hat{Z}_1 + \frac{\tau^3}{6} \hat{Z}_3 - L\hat{G}_1 L, \hat{\Psi}_{1,3} = \frac{1}{\tau} \hat{R}_{13}, \hat{\Psi}_{1,4} = \hat{P}_1 + \tau \hat{R}_{12}^T - (X_1 - Y_1)^T, \\ \hat{\Psi}_{1,5} &= L^T \hat{G}_2^T, \hat{\Psi}_{1,7} = -\frac{\hat{R}_{12}}{\tau} + \frac{2\hat{Z}_1}{\tau}, \hat{\Psi}_{2,2} = -(1 - \mu)\hat{R}_3, \hat{\Psi}_{2,6} = L^T \hat{G}_3^T, \hat{\Psi}_{3,3} = -\frac{1}{\tau} \hat{R}_{13}, \hat{\Psi}_{3,7} = \frac{1}{\tau} \hat{R}_{12}, \\ \hat{\Psi}_{4,4} &= \tau \hat{R}_{13} + \frac{\tau^2 \hat{Z}_1}{2} - 2X_1, \hat{\Psi}_{4,13} = AX_2, \hat{\Psi}_{4,14} = BX_2, \hat{\Psi}_{5,5} = -2\hat{G}_2 - \hat{G}_1, \hat{\Psi}_{5,12} = X_1^T C^T, \\ \hat{\Psi}_{6,6} &= -2\hat{G}_3, \hat{\Psi}_{6,12} = X_1^T D^T, \hat{\Psi}_{7,7} = -\frac{\hat{R}_{11}}{\tau} - \frac{2\hat{Z}_1}{\tau^2}, \hat{\Psi}_{8,8} = -\frac{6\hat{Z}_3}{\tau^3}, \hat{\Psi}_{9,9} = \hat{R}_4 + \tau \hat{R}_{21} - \frac{1}{\tau} \hat{R}_{23} - 2\hat{Z}_2 \\ &+ \frac{\tau^3 \hat{Z}_4}{6} - E\hat{G}_4 E, \hat{\Psi}_{9,11} = \frac{\hat{R}_{23}}{\tau}, \hat{\Psi}_{9,12} = \hat{P}_2 + \tau \hat{R}_{22}^T - (X_2 - Y_2)^T, \hat{\Psi}_{9,13} = E^T \hat{G}_5^T, \hat{\Psi}_{9,15} = -\frac{\hat{R}_{22}}{\tau} \\ &+ \frac{2\hat{Z}_2}{\tau}, \hat{\Psi}_{10,10} = -(1 - \mu)\hat{R}_4, \hat{\Psi}_{10,14} = E^T \hat{G}_6^T, \hat{\Psi}_{11,11} = -\frac{1}{\tau} \hat{R}_{23}, \hat{\Psi}_{11,15} = \frac{1}{\tau} \hat{R}_{22}, \hat{\Psi}_{12,12} = \tau \hat{R}_{23} \\ &+ \frac{\tau^2 \hat{Z}_2}{2} - 2X_2, \hat{\Psi}_{13,13} = -2\hat{G}_5 - \hat{G}_4, \hat{\Psi}_{14,14} = -2\hat{G}_6, \hat{\Psi}_{15,15} = -\frac{\hat{R}_{21}}{\tau} - \frac{2\hat{Z}_2}{\tau^2}, \hat{\Psi}_{16,16} = -\frac{6}{\tau^3} \hat{Z}_4 \end{aligned}$$

and the remaining terms of $\hat{\Psi}_{k \times l}$ are zero, then the slave system (7) is asymptotically synchronized with the master system (3) when the gain has type 1 perturbation. Moreover, the non-fragile observer-based control gain matrices K_1 and K_2 are then given by $K_1 = Y_1 X_1^{-1}$ and $K_2 = Y_2 X_2^{-1}$.

Proof. To derive the required results, we construct the Lyapunov–Krasovskii functional as follows:

$$V(t, e_x(t), e_y(t)) = \sum_{i=1}^5 V_i(t, e_x(t), e_y(t)), \quad (12)$$

where,

$$\begin{aligned} V_1(t, e_x(t), e_y(t)) &= e_x^T(t) P_1 e_x(t) + e_y^T(t) P_2 e_y(t), \\ V_2(t, e_x(t), e_y(t)) &= \int_{t-\tau(t)}^t e_x^T(s) R_3 e_x(s) ds + \int_{t-\tau(t)}^t e_y^T(s) R_4 e_y(s) ds, \\ V_3(t, e_x(t), e_y(t)) &= \int_{-\tau}^0 \int_{t+\theta}^t \eta_x^T(s) R_1 \eta_x(s) ds + \int_{-\tau}^0 \int_{t+\theta}^t \eta_y^T(s) R_2 \eta_y(s) ds, \\ V_4(t, e_x(t), e_y(t)) &= \int_{-\tau}^0 \int_{\theta_1}^0 \int_{t+\theta_2}^t \dot{e}_x^T(s) Z_1 \dot{e}_x(s) ds d\theta_2 d\theta_1 + \int_{-\tau}^0 \int_{\theta_1}^0 \int_{t+\theta_2}^t \dot{e}_y^T(s) Z_2 \dot{e}_y(s) ds d\theta_2 d\theta_1, \\ V_5(t, e_x(t), e_y(t)) &= \int_{-\tau}^0 \int_{\theta_1}^0 \int_{\theta_2}^0 \int_{t+\theta_3}^t e_x^T(s) Z_3 e_x(s) ds d\theta_3 d\theta_2 d\theta_1 + \int_{-\tau}^0 \int_{\theta_1}^0 \int_{\theta_2}^0 \int_{t+\theta_3}^t e_y^T(s) Z_4 e_y(s) ds d\theta_3 d\theta_2 d\theta_1, \end{aligned}$$

with $\eta_x^T(s) = [e_x(s) \quad \dot{e}_x(s)]^T$ and $\eta_y^T(s) = [e_y(s) \quad \dot{e}_y(s)]^T$. Let the infinitesimal operator $\mathcal{L}$ of $V(t, e(t))$ is defined as $\mathcal{L}V(t, e_x(t), e_y(t)) = \lim_{\Delta \rightarrow 0^+} \frac{1}{\Delta} \{ \mathbb{E}[V(e_{t+\Delta}) | x_t] - V(e_t) \}$. Now, we can calculate stochastic differential $\mathcal{L}V_i(t, e_x(t), e_y(t))$, $i = 1, 2, \dots, 5$ along the trajectories of the system (10), then we have

$$\mathbb{E}[\mathcal{L}V_1(t, e_x(t), e_y(t))] = \mathbb{E}[2e_x^T(t) P_1 \dot{e}_x(t) + 2e_y^T(t) P_2 \dot{e}_y(t)], \quad (13)$$

$$\begin{aligned} \mathbb{E}[\mathcal{L}V_2(t, e_x(t), e_y(t))] &= \mathbb{E}[e_x^T(t) R_3 e_x(t) - (1 - \mu) e_x^T(t - \tau(t)) R_3 e_x(t - \tau(t)) \\ &\quad + e_y^T(t) R_4 e_y(t) - (1 - \mu) e_y^T(t - \tau(t)) R_4 e_y(t - \tau(t))], \end{aligned} \quad (14)$$

$$\begin{aligned} \mathbb{E}[\mathcal{L}V_3(t, e_x(t), e_y(t))] &= \mathbb{E}[\tau e_x^T(t) R_{11} e_x(t) + 2\tau e_x^T(t) R_{12} e_x(t) + \tau e_x^T(t) R_{13} \dot{e}_x(t) - \int_{t-\tau}^t \eta_x^T(s) R_1 \eta_x(s) ds \\ &\quad + \tau e_y^T(t) R_{21} e_y(t) + 2\tau e_y^T(t) R_{22} e_y(t) + \tau e_y^T(t) R_{23} \dot{e}_y(t) - \int_{t-\tau}^t \eta_y^T(s) R_2 \eta_y(s) ds], \end{aligned} \quad (15)$$

$$\begin{aligned} \mathbb{E}[\mathcal{L}V_4(t, e_x(t), e_y(t))] &\leq \mathbb{E}\left[\frac{\tau^2}{2} \dot{e}_x^T(t) Z_1 \dot{e}_x(t) - \int_{-\tau}^0 \int_{t+\theta}^t \dot{e}_x^T(s) Z_1 \dot{e}_x(s) ds d\theta \right. \\ &\quad \left. + \frac{\tau^2}{2} \dot{e}_y^T(t) Z_2 \dot{e}_y(t) - \int_{-\tau}^0 \int_{t+\theta}^t \dot{e}_y^T(s) Z_2 \dot{e}_y(s) ds d\theta \right], \end{aligned} \quad (16)$$

$$\begin{aligned} \mathbb{E}[\mathcal{L}V_5(t, e_x(t), e_y(t))] &\leq \mathbb{E}\left[\frac{\tau^3}{6} e_x^T(t) Z_3 e_x(t) - \int_{-\tau}^0 \int_{\theta_1}^0 \int_{t+\theta_2}^t e_x^T(s) Z_3 e_x(s) ds d\theta_2 d\theta_1 \right. \\ &\quad \left. + \frac{\tau^3}{6} e_y^T(t) Z_4 e_y(t) - \int_{-\tau}^0 \int_{\theta_1}^0 \int_{t+\theta_2}^t e_y^T(s) Z_4 e_y(s) ds d\theta_2 d\theta_1 \right]. \end{aligned} \quad (17)$$

Now applying the Lemma 2.2, we have

$$- \int_{t-\tau}^t \eta_x^T(s) R_1 \eta_x(s) ds \leq \frac{-1}{\tau} \left(\int_{t-\tau}^t \eta_x(s) ds \right)^T \begin{bmatrix} R_{11} & R_{12} \\ R_{13} & \end{bmatrix} \left(\int_{t-\tau}^t \eta_x(s) ds \right), \quad (18)$$

$$- \int_{t-\tau}^t \eta_y^T(s) R_2 \eta_y(s) ds \leq \frac{-1}{\tau} \left(\int_{t-\tau}^t \eta_y(s) ds \right)^T \begin{bmatrix} R_{21} & R_{22} \\ R_{23} & \end{bmatrix} \left(\int_{t-\tau}^t \eta_y(s) ds \right), \quad (19)$$

$$- \int_{-\tau}^0 \int_{t+\theta}^t \dot{e}_x^T(s) Z_1 \dot{e}_x(s) ds d\theta \leq \frac{-2}{\tau^2} \left[\tau e_x^T(t) - \left(\int_{t-\tau}^t e_x(s) ds \right)^T \right] Z_1 \left[\tau e_x(t) - \left(\int_{t-\tau}^t e_x(s) ds \right) \right], \quad (20)$$

$$-\int_{-\tau}^0 \int_{t+\theta}^t \dot{e}_y^T(s) Z_2 \dot{e}_y(s) ds d\theta \leq \frac{-2}{\tau^2} \left[\tau e_y^T(t) - \left(\int_{t-\tau}^t e_y(s) ds \right)^T \right] Z_2 \left[\tau e_y(t) - \left(\int_{t-\tau}^t e_y(s) ds \right) \right], \quad (21)$$

$$-\int_{-\tau}^0 \int_{-\theta_1}^0 \int_{t+\theta_2}^t e_x^T(s) Z_3 e_x(s) ds d\theta_2 \sim d\theta_1 \leq \frac{-6}{\tau^3} \mathcal{G}_1^T Z_3 \mathcal{G}_1, \quad (22)$$

$$-\int_{-\tau}^0 \int_{-\theta_1}^0 \int_{t+\theta_2}^t e_y^T(s) Z_4 e_y(s) ds d\theta_2 \sim d\theta_1 \leq \frac{-6}{\tau^3} \mathcal{G}_2^T Z_4 \mathcal{G}_2, \quad (23)$$

where $\mathcal{G}_1 = \int_{-\tau}^0 \int_{-\theta_1}^0 \int_{t+\theta_2}^t e_x(s) ds d\theta_2 \sim d\theta_1$, $\mathcal{G}_2 = \int_{-\tau}^0 \int_{-\theta_1}^0 \int_{t+\theta_2}^t e_y(s) ds d\theta_2 \sim d\theta_1$. Then, for any matrices N_1 and N_2 with appropriate dimensions the following equations hold:

$$\begin{cases} \mathbb{E}[2\dot{e}_x^T(t) N_1 [(-I + K_1 + \alpha_a \Delta(t) K_1) e_x(t) + A f(e_y(t)) + B f(e_y(t - \tau(t))) - \dot{e}_x(t)]] = 0, \\ \mathbb{E}[2\dot{e}_y^T(t) N_2 [(-I + K_2 + \alpha_b \Delta(t) K_2) e_y(t) + C g(e_x(t)) + D g(e_x(t - \tau(t))) - \dot{e}_y(t)]] = 0. \end{cases} \quad (24)$$

Further, for positive diagonal matrices $G_1, G_2, G_3, G_4, G_5, G_6$ and using the assumption (A1), we have

$$e_x^T(t) L G_1 L e_x(t) - g^T(e_x(t)) G_1 g(e_x(t)) \geq 0, \quad (25)$$

$$2g^T(e_x(t)) G_2 (L e_x(t) - g(e_x(t))) \geq 0, \quad (26)$$

$$2g^T(e_x(t - \tau(t))) G_3 (L e_x(t - \tau(t)) - g(e_x(t - \tau(t)))) \geq 0, \quad (27)$$

$$e_y^T(t) S G_4 S e_y(t) - f^T(e_y(t)) G_4 f(e_y(t)) \geq 0, \quad (28)$$

$$2f^T(e_y(t)) G_5 (S e_y(t) - f(e_y(t))) \geq 0, \quad (29)$$

$$2f^T(e_y(t - \tau(t))) G_6 (S e_y(t - \tau(t)) - f(e_y(t - \tau(t)))) \geq 0. \quad (30)$$

Now combining (12)–(30), we have

$$\mathbb{E}\{\mathcal{LV}(s)\} \leq \mathbb{E}[\lambda^T(t)(\Psi + \Delta\Psi(t))\lambda(t)], \quad (31)$$

where $\Delta\Psi(t) = M_a F(t) M_b + M_b^T F^T(t) M_a^T + M_c F(t) M_d + M_d^T F^T(t) M_c^T$. Now by using the Schur complement, $\Delta\Psi(t)$ becomes,

$$\Delta\Psi(t) \leq \epsilon_1^{-1} M_a M_a^T + \epsilon_1 M_b^T M_b + \epsilon_2^{-1} M_c M_c^T + \epsilon_2 M_d^T M_d.$$

Then (31) becomes

$$\mathbb{E}\{\mathcal{LV}(s)\} \leq \mathbb{E}[\lambda^T(t) \tilde{\Psi} \lambda(t)], \quad (32)$$

where $\tilde{\Psi} = [\Psi_{k \times l}]_{16 \times 16} + \epsilon_1^{-1} M_a M_a^T + \epsilon_1 M_b^T M_b + \epsilon_2^{-1} M_c M_c^T + \epsilon_2 M_d^T M_d$, $\lambda^T(t) = \begin{bmatrix} e_x^T(t) & e_x^T(t - \tau(t)) & e_x^T(t - \tau) & e_x^T(t) & g^T(e_x(t)) \\ g^T(e_x(t - \tau(t))) & \int_{t-\tau}^t e_x^T(s) ds \left(\int_{-\tau}^0 \int_{-\theta_1}^0 \int_{t+\theta_2}^t e_x(s) ds d\theta_2 \sim d\theta_1 \right)^T & e_y^T(t) & e_y^T(t - \tau(t)) e_y^T(t - \tau) & e_y^T(t) & f^T(e_y(t)) & f^T(e_y(t - \tau(t))) \\ \int_{t-\tau}^t e_y^T(s) ds \left(\int_{-\tau}^0 \int_{-\theta_1}^0 \int_{t+\theta_2}^t e_y(s) ds d\theta_2 \sim d\theta_1 \right)^T \end{bmatrix}$.

$$\Psi_{1,1} = R_3 + \tau R_{11} - \frac{1}{\tau} R_{13} - 2\hat{Z}_1 + \frac{\tau^3}{6} Z_3 - L G_1 L, \Psi_{1,3} = -\frac{1}{\tau} R_{13}, \Psi_{1,4} = P_1 + \tau R_{12}^T + K_1^T N_1^T - N_1^T,$$

$$\Psi_{1,5} = L^T G_2^T, \Psi_{1,7} = -\frac{R_{12}}{\tau} + \frac{2Z_1}{\tau}, \Psi_{2,2} = -(1 - \mu) R_3, \Psi_{2,6} = L^T G_3^T, \Psi_{3,3} = -\frac{1}{\tau} R_{13}, \Psi_{3,7} = \frac{1}{\tau} R_{12},$$

$$\Psi_{4,4} = \tau R_{13} + \frac{\tau^2 Z_1}{2} - 2N_1, \Psi_{4,13} = N_1 A, \Psi_{4,14} = N_1 B, \Psi_{5,5} = -2G_2 - G_1, \Psi_{5,12} = C^T N_2^T, \Psi_{6,6} = -2G_3,$$

$$\Psi_{6,12} = D^T N_2^T, \Psi_{7,7} = -\frac{R_{11}}{\tau} - \frac{2Z}{\tau^2}, \Psi_{8,8} = -\frac{6}{\tau^3} Z_3, \Psi_{9,9} = R_4 + \tau R_{21} - \frac{1}{\tau} R_{23} - 2Z_2 + \frac{\tau^3}{6} Z_4 - E G_4 E,$$

$$\Psi_{9,11} = -\frac{1}{\tau} R_{23}, \Psi_{9,12} = P_2 + \tau R_{22}^T + K_2^T N_2^T - N_2^T, \Psi_{9,13} = E^T G_5^T, \Psi_{9,15} = -\frac{R_{22}}{\tau} + \frac{2Z_2}{\tau},$$

$$\Psi_{10,10} = -(1 - \mu) R_4, \Psi_{10,14} = E^T G_6^T, \Psi_{11,11} = -\frac{1}{\tau} R_{23}, \Psi_{11,15} = \frac{1}{\tau} R_{22}, \Psi_{12,12} = \tau R_{23} + \frac{\tau^2 Z_2}{2} - 2N_2,$$

$$\Psi_{13,13} = -2G_5 - G_4, \Psi_{14,14} = 2G_6, \Psi_{15,15} = -\frac{R_{21}}{\tau} - \frac{2Z_2}{\tau^2}, \Psi_{16,16} = -\frac{6}{\tau^3} Z_4, M_a = [0_{n \times 3n} \ N_1^T M_1^T \ 0_{n \times 12n}],$$

$$M_b = [\alpha_a H_1 \ 0_{n \times 15n}], M_c = [0_{n \times 11n} \ N_2^T M_2^T \ 0_{n \times 4n}], M_d = [0_{n \times 8n} \ \alpha_b H_2 \ 0_{n \times 7n}].$$

$$\begin{aligned}\tilde{\Pi}_{9,11} &= \frac{R_{23}}{\tau}, \tilde{\Pi}_{9,12} = P_2 + \tau R_{22}^T + K_2^T N_2^T - N_2^T, \tilde{\Pi}_{9,13} = E^T G_5^T, \tilde{\Pi}_{9,15} = -\frac{R_{22}}{\tau} + \frac{2Z_2}{\tau}, \\ \tilde{\Pi}_{10,10} &= -(1-\mu)R_4, \tilde{\Pi}_{10,14} = E^T G_6^T, \tilde{\Pi}_{11,11} = -\frac{R_{23}}{\tau}, \tilde{\Pi}_{11,15} = \frac{R_{22}}{\tau}, \tilde{\Pi}_{12,12} = \tau R_{23} + \frac{\tau^2 Z_2}{2} - 2N_2, \\ \tilde{\Pi}_{13,13} &= -2G_5 - G_4, \tilde{\Pi}_{14,14} = -2G_6, \tilde{\Pi}_{15,15} = -\frac{R_{21}}{\tau} - \frac{2Z_2}{\tau^2}, \tilde{\Pi}_{16,16} = \frac{-6Z_4}{\tau^3}, \tilde{M}_a = [0_{n \times 3n} \ (N_1 K_1)^T M_1^T \ 0_{n \times 12n}], \\ \tilde{M}_b &= [\alpha_a H_1 \ 0_{n \times 15n}], \tilde{M}_c = [0_{n \times 11n} \ (N_2 K_2)^T M_2^T \ 0_{n \times 4n}], \tilde{M}_d = [0_{n \times 8n} \ \alpha_b H_2 \ 0_{n \times 7n}].\end{aligned}$$

As similar to the conclusion of [Theorem 3.1](#), we have $\mathcal{L}V(t) < 0$. Thus we conclude that by Lyapunov stability theory, the slave system (7) can globally asymptotically synchronize with the master system (6), which completes the proof. $\square$

Remark 3.3. It is worth pointing out that the non-fragile controller of randomly occurring controller gain fluctuation is interesting but has not attracted enough attention. Thus, the theoretical results proposed in this paper enrich the study on synchronization of memristive BAM neural networks with time-varying delays.

Remark 3.4. It should be pointed that [34], the non-fragile synchronization problem has been investigated for complex networks with time-varying delay and missing control signal. In this paper, the randomness of controller gain fluctuations has been taken into account for the first time to study the non-fragile synchronization problem for memristive BAM neural networks with two types of perturbations. As it is well known that the complex networks and memristive neural networks are different types with their own practical usages. Further, extending present results to NNs with diverse property of time delays [43] and different types of time delays will be our future topics of research.

4. Numerical simulation

Now, we perform some numerical simulations to demonstrate the effectiveness and applicability of the proposed results. Consider a two-dimensional memristor-based BAM neural network system (3) with two types of gain variations. First let us consider [Theorem 3.1](#) with additive gain variations as we introduced in Type 1.

Example for Type 1 perturbation: Let us assume the weight parameters are satisfying

$$\begin{aligned}a_{11}(x_1(t)) &= \begin{cases} -0.45, & |x_1(t)| < 1 \\ 0.45, & |x_1(t)| > 1 \end{cases}, \quad a_{12}(x_1(t)) = \begin{cases} -0.5, & |x_1(t)| < 1 \\ 0.5, & |x_1(t)| > 1 \end{cases}, \\ a_{21}(x_2(t)) &= \begin{cases} -0.1, & |x_2(t)| < 1 \\ 0.1, & |x_2(t)| > 1 \end{cases}, \quad a_{22}(x_2(t)) = \begin{cases} -2.3, & |x_2(t)| < 2 \\ 2.3, & |x_2(t)| > 2 \end{cases}, \\ b_{11}(x_1(t)) &= \begin{cases} -0.54, & |x_1(t)| < 1 \\ 0.54, & |x_1(t)| > 1 \end{cases}, \quad b_{12}(x_2(t)) = \begin{cases} -0.8, & |x_2(t)| < 1 \\ 0.8, & |x_2(t)| > 1 \end{cases}, \\ b_{21}(x_1(t)) &= \begin{cases} -0.6, & |x_1(t)| < 1 \\ 0.6, & |x_1(t)| > 1 \end{cases}, \quad b_{22}(x_2(t)) = \begin{cases} -1.2, & |x_2(t)| < 2 \\ 1.2, & |x_2(t)| > 2 \end{cases}, \\ c_{11}(y_1(t)) &= \begin{cases} -0.8, & |y_1(t)| < 1 \\ 0.8, & |y_1(t)| > 1 \end{cases}, \quad c_{12}(y_1(t)) = \begin{cases} -0.6, & |y_1(t)| < 1 \\ 0.6, & |y_1(t)| > 1 \end{cases}, \\ c_{21}(y_2(t)) &= \begin{cases} -0.5, & |y_2(t)| < 1 \\ 0.5, & |y_2(t)| > 1 \end{cases}, \quad c_{22}(y_2(t)) = \begin{cases} -0.8, & |y_2(t)| < 1 \\ 0.8, & |y_2(t)| > 1 \end{cases}, \\ d_{11}(y_1(t)) &= \begin{cases} -0.5, & |y_1(t)| < 1 \\ 0.5, & |y_1(t)| > 1 \end{cases}, \quad d_{12}(y_2(t)) = \begin{cases} -0.19, & |y_2(t)| < 1 \\ 0.19, & |y_2(t)| > 1 \end{cases}, \\ d_{21}(y_1(t)) &= \begin{cases} -0.51, & |y_1(t)| < 1 \\ 0.51, & |y_1(t)| > 1 \end{cases}, \quad d_{22}(y_2(t)) = \begin{cases} -0.5, & |y_2(t)| < 1 \\ 0.5, & |y_2(t)| > 1 \end{cases}.\end{aligned}$$

Then there exist,

$$\begin{aligned}A &= \begin{bmatrix} -0.45 & 0.5 \\ -0.1 & -2.3 \end{bmatrix}, \quad B = \begin{bmatrix} 0.54 & -0.8 \\ 0.6 & 1.2 \end{bmatrix}, \quad C = \begin{bmatrix} 0.8 & 0.6 \\ 0.5 & 0.8 \end{bmatrix}, \quad D = \begin{bmatrix} 0.5 & -0.19 \\ 0.51 & 0.5 \end{bmatrix}, \quad L = E = \text{diag}\{0.8, 0.8\}, \\ M_1 &= \begin{bmatrix} -0.4 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}, \quad M_2 = \begin{bmatrix} -0.3 & 0.9 \\ 0.15 & -1.5 \end{bmatrix}, \quad H_1 = \begin{bmatrix} -0.5 & -0.6 \\ -0.5 & 0.4 \end{bmatrix}, \quad H_2 = \begin{bmatrix} 0.5 & 0.5 \\ -0.2 & 0.5 \end{bmatrix},\end{aligned}$$

and the nonlinear activation functions are taken as $\tilde{g}_i(x_i) = \tanh(0.8x_i)$, $\tilde{f}_j(y_j) = \tanh(0.8y_j)$, $i, j = 1, 2$. For the given values $\tau = 0.9$, $\mu = 0.5$, $\alpha_a = 0.5$, $\alpha_b = 0.7$ and with the above parameters, by MATLAB LMI toolbox the LMI (11) holds. According to [Theorem 3.1](#), we can conclude that the system (6) will synchronize with its response system (7) under non-fragile observer based

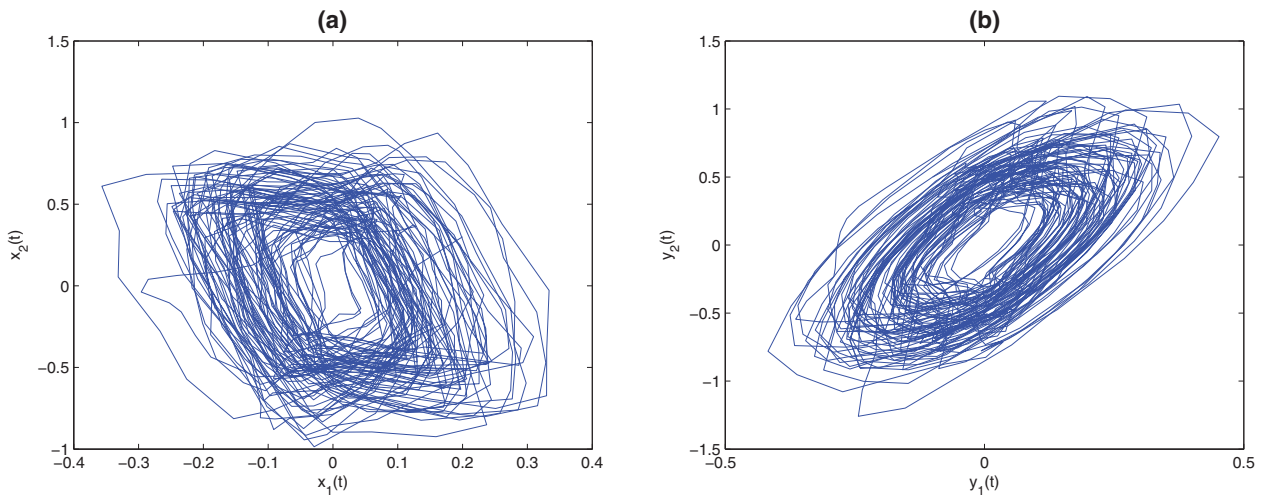


Fig. 1. The chaotic behavior of the master system (6).

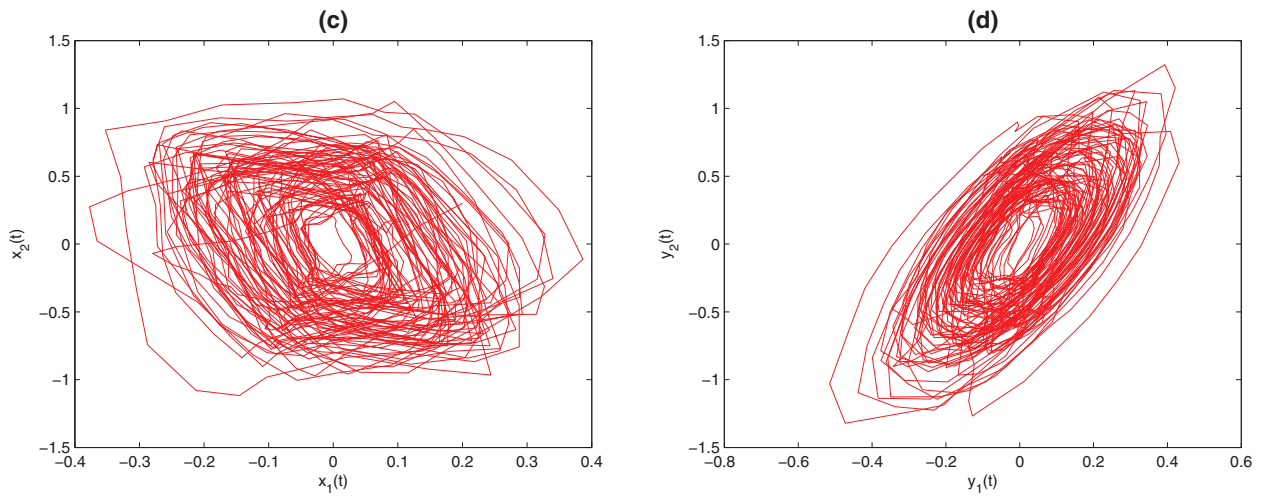


Fig. 2. The chaotic behavior of the slave system (7).

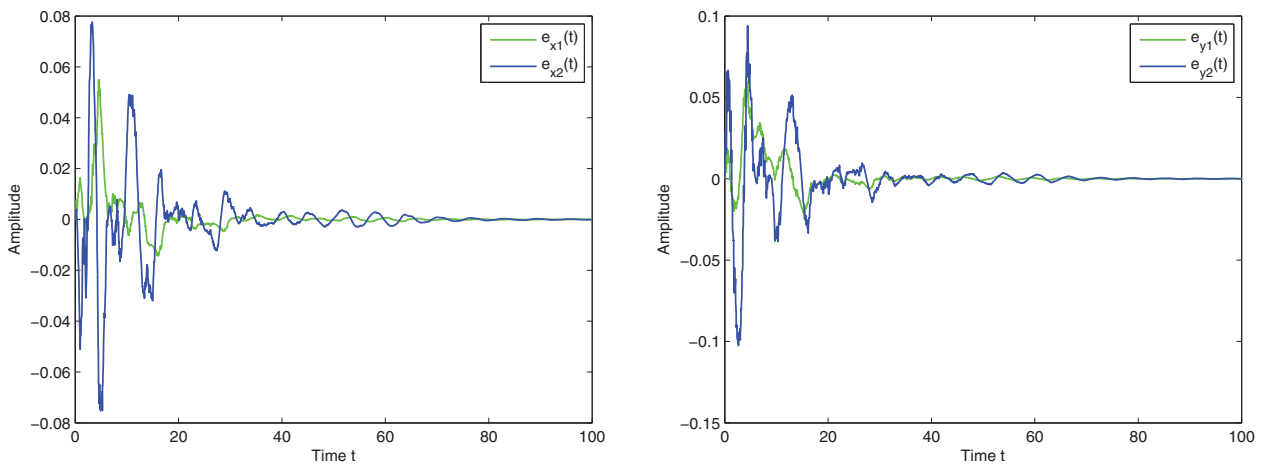


Fig. 3. Response of the error system (10).

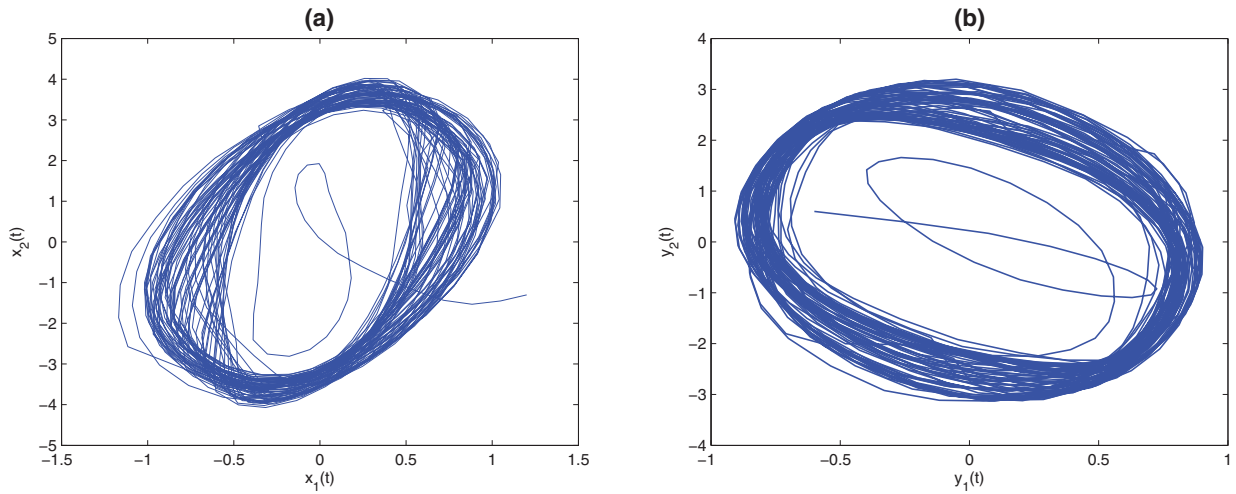


Fig. 4. The chaotic attractor of the master system (6).

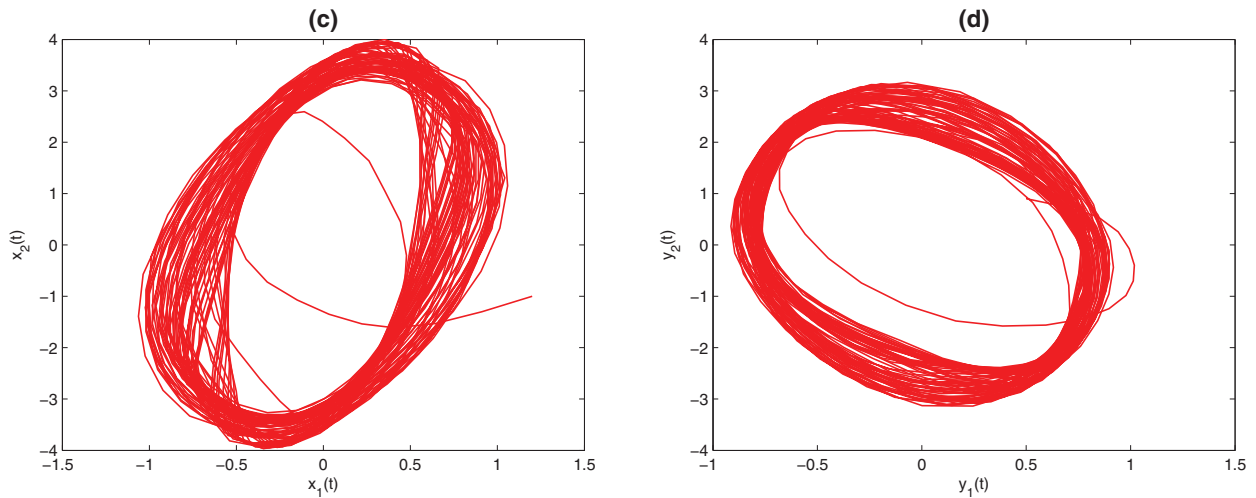


Fig. 5. The chaotic attractor of the slave system (7).

controller with gain matrices K_1 and K_2 given by

$$K_1 = \begin{bmatrix} 0.3819 & 0.0434 \\ 0.1772 & 0.4925 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 0.3803 & -0.0705 \\ 0.0552 & 1.0729 \end{bmatrix}.$$

The chaotic behavior of the considered master system and slave system with initial conditions $x(s) = [0.2 \ 0.3]^T$, $y(s) = [-0.1 \ 0.5]^T$ are shown in Figs. 1 and 2, respectively. Also, Fig. 3(a) and (b) depicts the synchronization error of the state variables $e_x(t)$ and $e_y(t)$ between the drive system and the response system with the non-fragile controllers, respectively. From the simulation figures, we can conclude that the slave system (7) can globally asymptotically synchronized with the master system (6). Hence the numerical simulations has confirmed our theoretical results.

Now we will prove, the obtained robust non-fragile controller guarantees the synchronization of the closed-loop system in spite of the given multiplicative gain variations via simulations.

Example for Type 2 perturbation: Consider the two-dimensional memristor-based BAM neural network system (3) with the following parameters

$$a_{11}(x_1(t)) = \begin{cases} -0.41, & |x_1(t)| < 1 \\ 0.41, & |x_1(t)| > 1 \end{cases}, \quad a_{12}(x_1(t)) = \begin{cases} -0.5, & |x_1(t)| < 1 \\ 0.5, & |x_1(t)| > 1 \end{cases},$$

$$a_{21}(x_2(t)) = \begin{cases} -0.4, & |x_2(t)| < 1 \\ 0.4, & |x_2(t)| > 1 \end{cases}, \quad a_{22}(x_2(t)) = \begin{cases} -2.5, & |x_2(t)| < 1 \\ 2.5, & |x_2(t)| > 1 \end{cases},$$

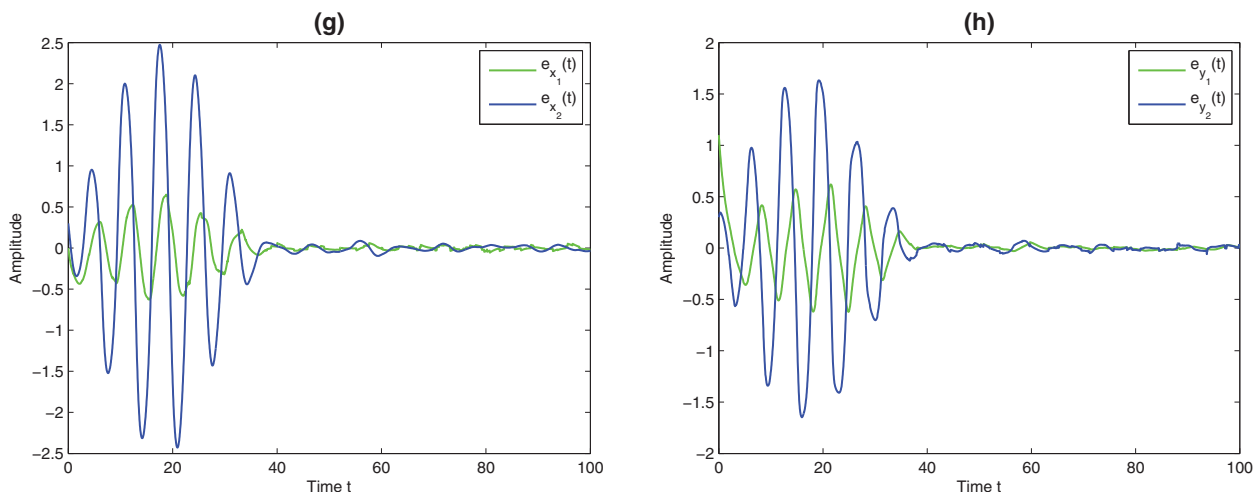


Fig. 6. Response of the error system according to Theorem 3.2.

$$\begin{aligned}
 b_{11}(x_1(t)) &= \begin{cases} -0.4, & |x_1(t)| < 1 \\ 0.4, & |x_1(t)| > 1 \end{cases}, & b_{12}(x_2(t)) &= \begin{cases} -0.8, & |x_2(t)| < 1 \\ 0.8, & |x_2(t)| > 1 \end{cases}, \\
 b_{21}(x_1(t)) &= \begin{cases} -0.5, & |x_1(t)| < 1 \\ 0.5, & |x_1(t)| > 1 \end{cases}, & b_{22}(x_2(t)) &= \begin{cases} -1, & |x_2(t)| < 1 \\ 1, & |x_2(t)| > 1 \end{cases}, \\
 c_{11}(y_1(t)) &= \begin{cases} -0.5, & |y_1(t)| < 1 \\ 0.5, & |y_1(t)| > 1 \end{cases}, & c_{12}(y_1(t)) &= \begin{cases} -0.5, & |y_1(t)| < 1 \\ 0.5, & |y_1(t)| > 1 \end{cases}, \\
 c_{21}(y_2(t)) &= \begin{cases} -0.6, & |y_2(t)| < 1 \\ 0.6, & |y_2(t)| > 1 \end{cases}, & c_{22}(y_2(t)) &= \begin{cases} -0.8, & |y_2(t)| < 1 \\ 0.8, & |y_2(t)| > 1 \end{cases}, \\
 d_{11}(y_1(t)) &= \begin{cases} -0.51, & |y_1(t)| < 1 \\ 0.51, & |y_1(t)| > 1 \end{cases}, & d_{12}(y_2(t)) &= \begin{cases} -0.19, & |y_2(t)| < 1 \\ 0.19, & |y_2(t)| > 1 \end{cases}, \\
 d_{21}(y_1(t)) &= \begin{cases} -0.51, & |y_1(t)| < 1 \\ 0.51, & |y_1(t)| > 1 \end{cases}, & d_{22}(y_2(t)) &= \begin{cases} -1.3, & |y_2(t)| < 1 \\ 1.3, & |y_2(t)| > 1 \end{cases}.
 \end{aligned}$$

Then there exist,

$$\begin{aligned}
 A &= \begin{bmatrix} -0.41 & 0.5 \\ 0.4 & -2.5 \end{bmatrix}, \quad B = \begin{bmatrix} -0.4 & -0.8 \\ 0.5 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 0.5 & -0.5 \\ 0.6 & 0.8 \end{bmatrix}, \quad D = \begin{bmatrix} 0.51 & -0.19 \\ 0.51 & 1.3 \end{bmatrix}, \quad M_1 = \begin{bmatrix} -0.4 & -0.5 \\ -0.7 & -0.6 \end{bmatrix}, \\
 M_2 &= \begin{bmatrix} -0.3 & 0.9 \\ 0.15 & -1.5 \end{bmatrix}, \quad H_1 = \begin{bmatrix} -0.5 & -0.6 \\ -0.5 & 0.4 \end{bmatrix}, \quad H_2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.5 \end{bmatrix}, \quad L = E = \text{diag}\{0.6, 0.6\}
 \end{aligned}$$

and the nonlinear activation functions are taken as $\tilde{g}_i(x_i) = \tanh(0.6x_i)$, $\tilde{f}_j(y_j) = \tanh(0.6y_j)$, $i, j = 1, 2$. For the given values $\tau = 1$, $\mu = 0.45$, $\alpha_a = 0.02$, $\alpha_b = 0.02$ and with the above parameters, by solving the LMI (33), the non-fragile observer based control gain matrices K_1 and K_2 can be obtained as

$$K_1 = \begin{bmatrix} 0.3840 & 0.0338 \\ 0.0063 & 0.5177 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 0.4010 & 0.0546 \\ 0.0523 & 0.7597 \end{bmatrix}.$$

By taking the initial conditions $x(s) = [1.2 \quad -1.3]^T$ and $y(s) = [-0.6 \quad 0.6]^T$, the chaotic behaviors of master system and slave system are given in Figs. 4 and 5, respectively. Further, Fig. 6(a) and (b) have shown the synchronization error of the state variables $e_x(t)$ and $e_y(t)$, respectively for the controller with Type 2 perturbations. It is easy to seen from the simulation results and by Theorem 3.2 that the slave system (7) asymptotically synchronized with the master system (6). The numerical simulations reveal the effectiveness of the obtained results perfectly.

5. Conclusions

In this paper, we investigated a preliminary memristor-based BAM neural network model and some sufficient conditions to guarantee the non-fragile synchronization is derived. By constructing suitable LKF and using LMI approach with help of Jensens inequality, we derived several conditions to guarantee the non-fragile synchronization of the MBAMNNs with time delays. The non-fragile controller is designed by using the master–slave synchronization concept; the results are obtained for two types of randomly occurring controller gain fluctuations. Finally, a numerical example has been demonstrated to show the validity and potentiality of the developed theoretical results.

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